



Tracing Water Across Worlds: Water Distribution in Brown Dwarf Atmospheres

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Introduction

Brown dwarfs are substellar objects considered “too big” to be planets but “not big enough” to sustain hydrogen fusion, like a star [1]. These objects belong to a unique class that bridges the gap between stars and planets; studying them is crucial for further understanding the composition and evolution of stellar and planetary systems. In our study, **we investigate the distribution of water in the atmospheres of brown dwarfs**—specifically, how water absorption correlates with their poles, equator, and tilt (a property known as inclination). We hypothesized the equator could have less water absorption than at the poles [2].

Methods

To explore this relationship, we investigated the spectra of 10 different brown dwarfs, using 10 from the Spitzer space telescope and 1 from JWST [3]. First, we read the data in our jupyter notebook. Then, we measure the amount of water in the atmosphere, referred to as the water index (Figure 1; [4]). Afterward, we measured uncertainties for the water index using our uncertainties in our observations. Lastly, **we investigate the correlation between water index and the angle of viewing** by creating a graph to visualize Water Index vs. Inclination.

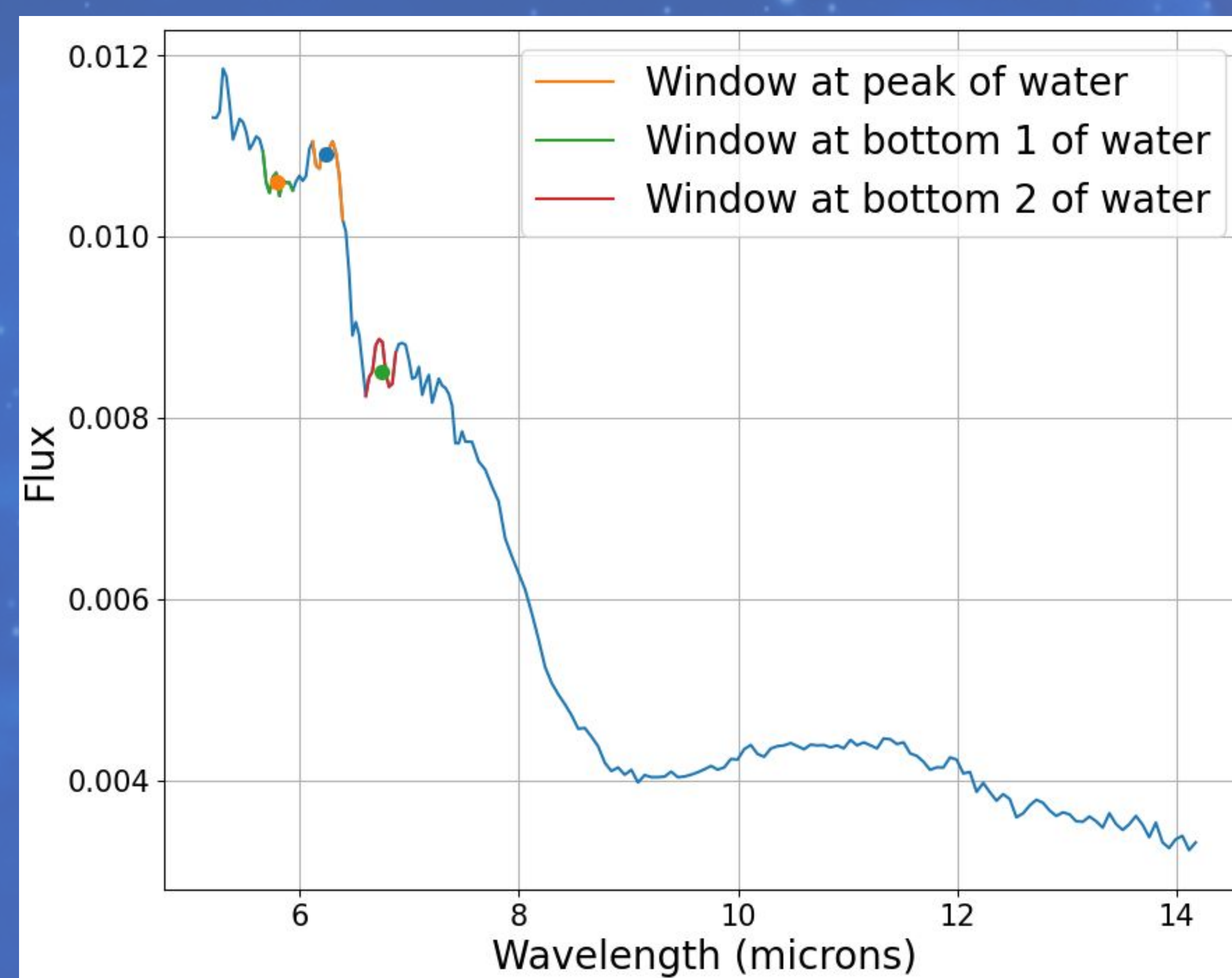


Figure 1: Measuring the amount of water in the spectra.

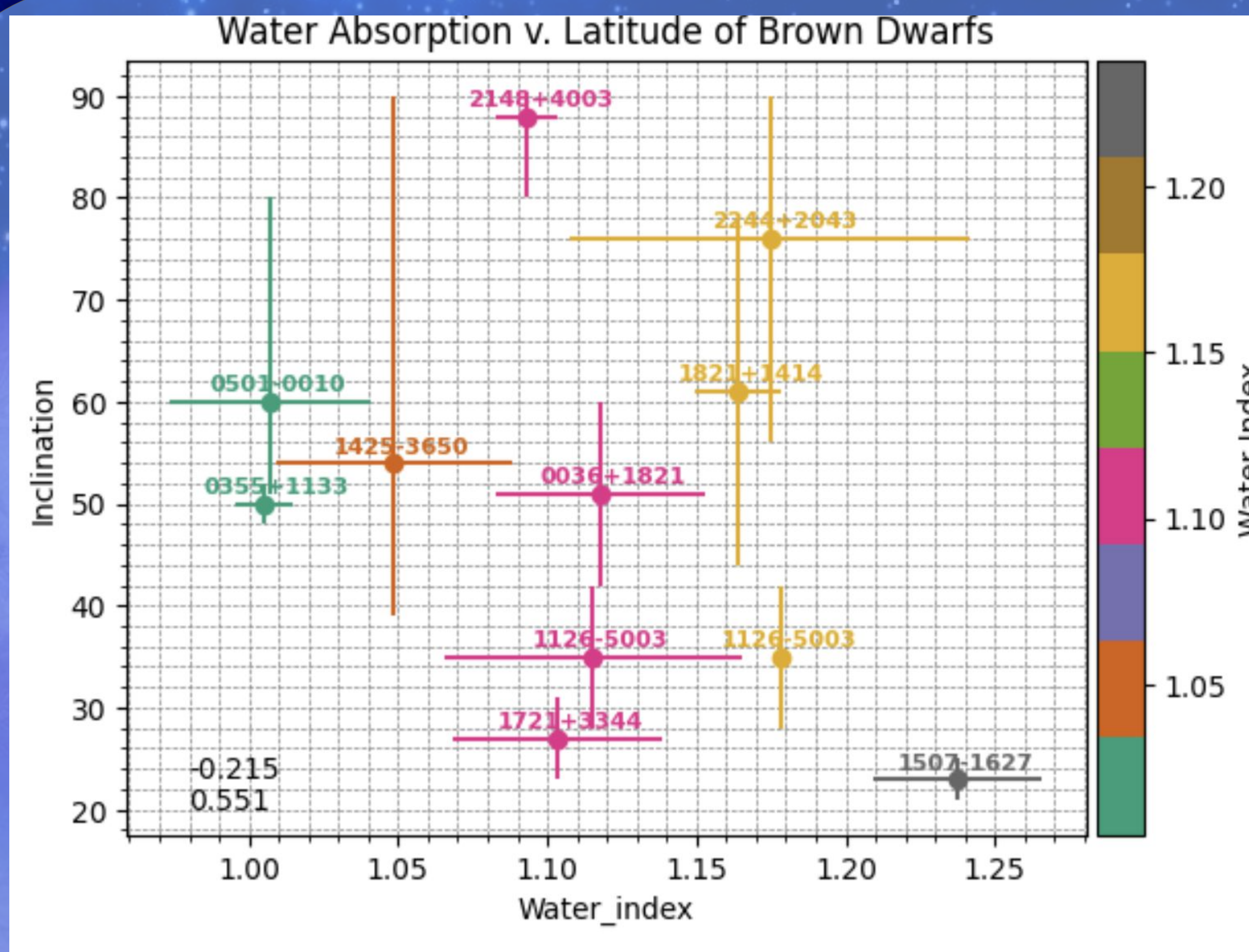
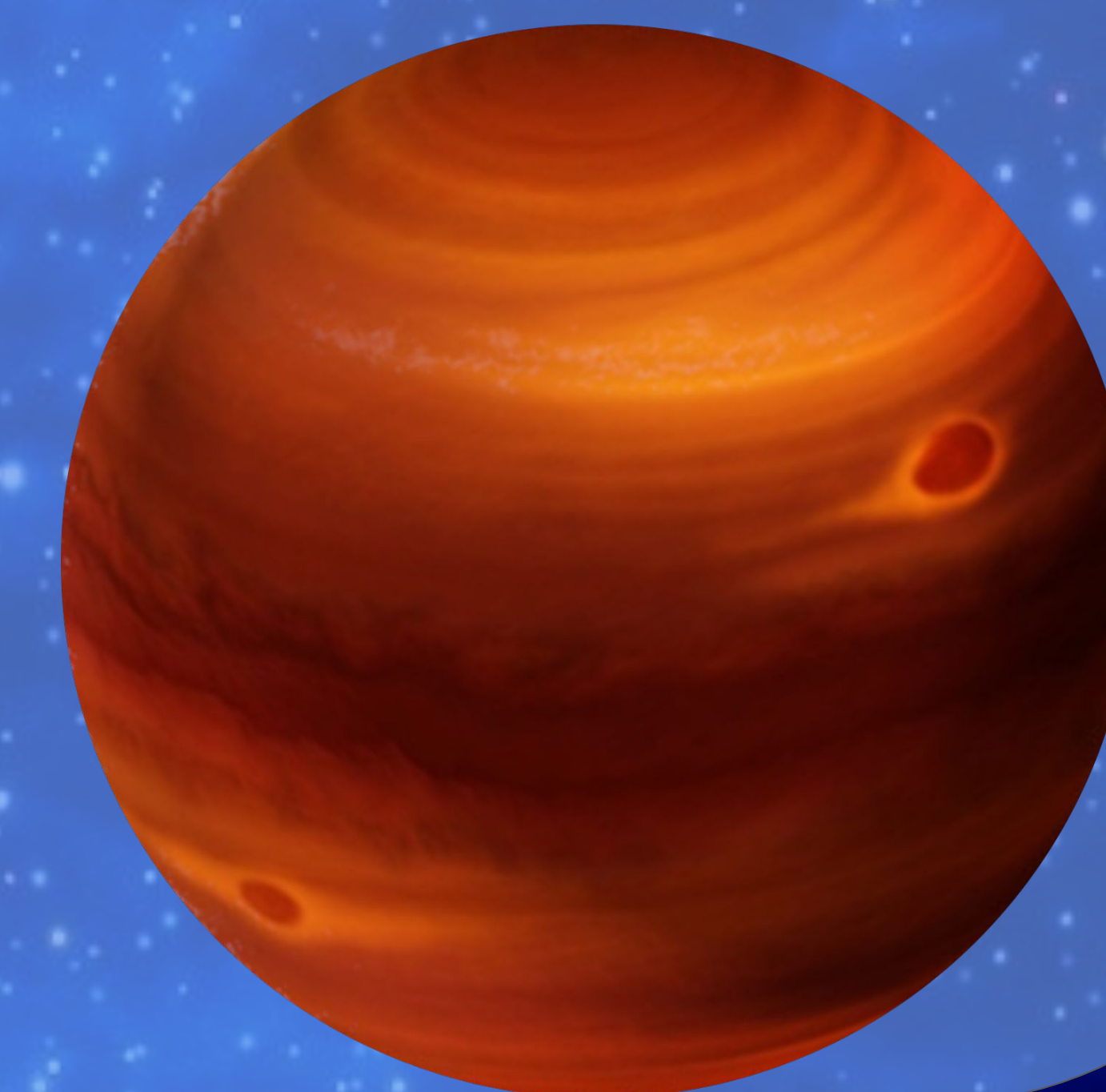


Figure 2: Relation between inclination and water index.

Results

We obtained our water index measurements from 10 brown dwarfs. We found no clear correlation between the water index and the inclination angle of the brown dwarfs in our sample, as seen by the scatter plot in Figure 2. This conclusion is supported by our calculated Pearson correlation coefficient of -0.215, which indicates a very weak or essentially null relationship between these two variables [5].



Discussion

Our results indicate that viewing a brown dwarf equator-on vs. pole-on will not show a significant difference in the amount of water detected. The results challenge our initial expectations that the amount of water should correlate with inclinations. However, we note that our graph has large uncertainties, so we need higher quality data to further investigate any potential correlations (e.g. Figure 3). The methodology used here is valuable, as it can be applied to other observations and improve our understanding of other exoplanets.

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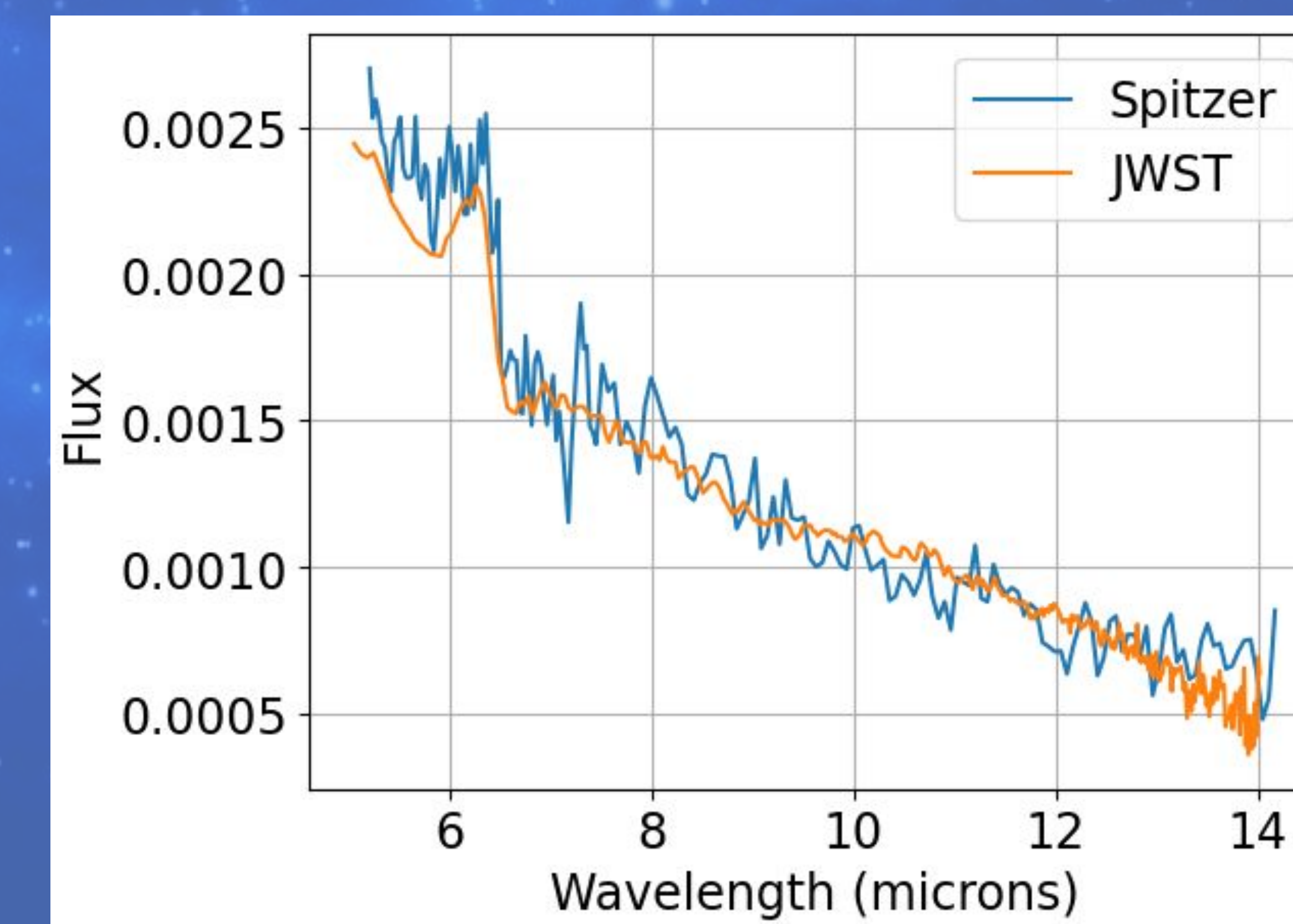


Figure 3: Comparison of Spitzer versus JWST data.

References

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